

Interspecies Communication App



Mobile App Prototype

Biology



3 hours 45 min



Grades 9-10



Student-led

The Point

The field of artificial intelligence (AI) and natural language processing will transform our understanding of non-human communication. How can AI enhance or facilitate interspecies interactions?

Objectives

- Explain how AI is being utilized to decode animal communication.
- Design a mobile app prototype for interspecies communication.
- Identify the potential benefits and limitations of using AI to facilitate communication between humans and animals.

Materials

- Internet-enabled devices for students
- Students will need to create an account on [Canva.com](https://www.canva.com)
- Access to the [Student Guide](#) for students
- Access to the [mobile app prototype template](#) for students

Activities

Make sure students have access to the [Student Guide](#) and [Canva.com](https://www.canva.com).

1. **Discuss** the warmup question, "Imagine you could talk to any animal in the world. Which animal would you choose and what would you say to it?" (5 min.)
2. **Review** the [AI in 5 Minutes](#) website page with students. (5 min.)
 - **Ask:** "Can you think of an example of how AI can help us learn more about how animals communicate with each other and with us?"
3. **Give** students time to complete the "Decoding animal communication" section in the student guide. (15 min.)
4. **Review** the vocabulary check to ensure students know the definition of interspecies and interspecies communication. (5 min.)
5. **Give** students time to complete the "Interspecies communication" section in the student guide. (10 min.)
6. **Discuss** student responses to the "Interspecies communication" section, use their prior experience as a segue to introduce the mobile app project. (5 min.)
 - **Example Segue Script** shared on the next page...

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Activities, continued

- **Example Segue Script:** *Think about the ways you've communicated with animals and how AI could enhance or facilitate those interactions. Your task is to design a mobile app that enables cooperative communication between humans and animals, using AI. Let your creativity flow and venture into the realm of speculative AI – imagining concepts that may not be a reality now but could come to life with future developments in AI.*
- 5. **Direct** students to read about "Speculative AI" and complete the "Brainstorm ideas" section in the student guide. **(20 min.)**
- 6. **Debrief** the "Brainstorm ideas" section activity with students, asking them how it went and if they have any questions. **(5 min.)**
 - If students are struggling to brainstorm ideas for their app, you can use the following series of questions to help guide them:
 - What types of animals are you most interested in communicating with? (e.g., pets, wild animals, marine life, etc.)
 - What specific communication barriers between humans and animals do you want to address? (e.g., understanding emotions, training, detecting distress signals, etc.)
 - How can AI be used to interpret or facilitate communication between humans and animals? (e.g., analyzing sounds, recognizing body language, identifying patterns in behavior, etc.)
 - What existing technologies or solutions can you draw inspiration from? (e.g., pet training apps, wildlife observation tools, the Earth Species Project, etc.)
 - How would your app enhance the user's experience or improve human-animal interactions? (e.g., better understanding of animal behavior, fostering empathy, improved training, etc.)
- 7. **Give** students time to complete the "Create your mobile app prototype" section in the student guide. **(120 min.)**
 - If students are struggling with using Canva.com, you can point them to the Canva video demonstration linked in the student guide (page 6).
- 8. **Give** students time to complete the "Get ready to pitch!" section in the student guide. **(20 min.)**
- 9. **Facilitate** student presentations or a gallery walk so students can share their prototype pitches with their peers. **(15 min.)**

